

COMSATS University Islamabad (Lahore Campus)

Department of Computer Engineering

Member A Sprint Plan

Platform, MQTT & Edge Deployment Lead

5 Months · 5 Sprints — Your Tasks Only

Master timeline: `.sprint/centralized.sprint_plan.tex`

Spring–Fall 2026

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1 Your Role

Your Scope Only

You own: Full MQTT pipeline (publisher → broker → bridge → consumers), Jetson deploy, 5 FastAPI services, PostgreSQL/MinIO/Redis, LangGraph agent, synthetic data, React dashboard, Docker/CI/CD/Grafana, event simulator.

Not Your Responsibility

Member B owns (do not implement): YOLO training, compliance algorithms, TensorRT model creation, annotation. You **deploy** B's merged PRs and wrap output in your MQTT publisher.

2 Sprint 1 (Weeks 1–4): Foundation & Transport

Your focus — Repo

Week 1–2

- Initialize `smart-surveillance/` monorepo; directory structure per architecture plan
- `infra/docker-compose.yml` skeleton; `.env.example`
- Draft `contracts/events.py` (Pydantic); share with Member B
- Dev environment: Python 3.11+, Node 20+, Docker Compose v2
- GitHub: branch rules, first PR, Issues board with `sprint-1` label

Week 3–4

- Mosquitto (:1883) + Redis (:6379) in Compose
- `transport/mqtt_redis_bridge/` — subscribe `ppe/#`, XADD to streams
- `transport/event_simulator/` — contract-valid test events
- Streams: `ppe:alerts`, `ppe:metrics`, `ppe:heartbeat`
- Document MQTT topic map in `contracts/README.md`

Your Sprint 1 done when:

- `docker compose up` starts Mosquitto + Redis + bridge + simulator
- Events visible via `XRANGE ppe:alerts - + COUNT 5`
- Prerequisites checklist signed (`.prerequisite/MEMBER_A_PREREQUISITES.md`)

Collaboration Touchpoint

Week 4 with Member B: Sign frozen contract (M0). Review B's handoff dict draft; confirm your publisher can serialize it.

3 Sprint 2 (Weeks 5–8): Backend Core

Your focus — Database + Alert + Metrics services

Week 5–6

- PostgreSQL + TimescaleDB; schema: `violations`, `compliance_metrics`, `zone_rules`
- MinIO `evidence/` bucket
- Alert Service (:8001): consumer group `alert-svc`, dedup TTL, DB writer
- GET `/api/alerts?zone=&priority=&from=&to=`

Week 7–8

- Metrics Service (:8002): hypertable writer, rolling aggregator, zone risk scorer
- GET `/api/metrics/compliance`, `/api/metrics/trend`, `/api/metrics/zone-risk`
- Redis pub/sub broadcast after alert processing (prep for WebSocket)
- Integration tests: simulator → DB row appears

Your Sprint 2 done when: M1 — violations and metrics queryable via REST from simulator events.

Collaboration Touchpoint

Week 8: Integration session. Optionally ingest B's sample handoff JSON via test script (publish to MQTT yourself).

4 Sprint 3 (Weeks 9–12): Services, Synthetic Data & Agent Start

Your focus — WebSocket

Week 9–10

- WebSocket Gateway (:8004): `/ws/live`, connection manager
- Reporting Service (:8003): ReportLab PDF, `POST /api/reports/generate`
- Synthetic engine: `synthetic/scenarios/90day_baseline.yaml`, run seed script

Week 11–12

- Agent Service (:8005): LangGraph planner + executor skeleton
- Tools: `query_violations`, `get_compliance_trend`, `get_zone_risk_score`
- `POST /api/agent/query` (basic chat)
- OpenAPI docs on all services

Your Sprint 3 done when: M2 — synthetic 90-day data in DB; WebSocket pushes simulator alerts; B's PR merged (you review, not write CV).

Collaboration Touchpoint

Week 12: Test harness reads B's handoff dict → your publisher → full stack. No changes to B's `edge/` code without B's PR.

5 Sprint 4 (Weeks 13–16): Dashboard & Jetson

Your focus — React UI

Week 13–14

- React + Vite + Zustand; WebSocket client
- Pages: Dashboard, Alerts, Analytics, Agent chat, Reports, Zone config
- Compliance gauge, alert feed, charts (Chart.js)
- Zone config: `requires_helmet`, `requires_vest` toggles

Week 15–16

- `edge/mqtt_publisher/` — wraps B's compliance output, publishes to Mosquitto
- Flash Jetson; deploy B's `edge/` + your publisher
- Agent: RAG (pgvector), reactive mode on CRITICAL, proactive cron
- Disable simulator when Jetson pipeline verified

Your Sprint 4 done when: M3 — live helmet-off demo on Jetson updates dashboard in real time.

Collaboration Touchpoint

Weeks 15–16: Member B on standby to explain CV thresholds. You own all Jetson SSH, broker IP, and restart during demo.

6 Sprint 5 (Weeks 17–20): DevOps & Demo

Your focus — Full stack**Week 17–18**

- Complete `docker-compose.yml` — all 12+ services, health checks
- GitHub Actions: lint → pytest → docker integration → build
- Prometheus `/metrics` on every FastAPI service

Week 19–20

- Grafana: System Health + Safety KPIs dashboards
- Rehearse 12-minute demo (see centralized plan)
- Platform chapters of FYP report; API documentation
- Tag release `v1.0-demo`

Your Sprint 5 done when: M4 — `docker compose up` demo flawless; CI green on main.

Collaboration Touchpoint

Week 20: Joint final demo with Member B. You operate stack + Jetson; B explains CV if supervisor asks.

7 Your Master Checklist

MQTT publisher, broker, bridge, stream consumers

Event simulator (Sprints 1–3)

Alert, Metrics, Reporting, WebSocket, Agent services

PostgreSQL + TimescaleDB + MinIO

Synthetic 90-day seed

React dashboard (6 pages)

Jetson flash, deploy, demo operation

Docker Compose + CI/CD + Grafana

Weekly sprint logs + GitHub Issues closed per sprint